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AI washing: Strategic disclosure and backlash

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ABSTRACT

As firms increasingly exaggerate artificial intelligence (AI) adoption in disclosures, we analyze market responses to unsubstantiated AI claims. Using BERT-based text classification and AI patent data for U.S. firms (2018–2023), we find that such narratives initially attract investors but ultimately yield negative market reactions and sustained underperformance. Results demonstrate market penalties for AI overclaiming through both short-term investor responses and long-term operational outcomes, extending narrative economics to emerging technologies. The study highlights that while narratives can mobilize attention, markets ultimately punish rhetoric that outpaces implementation.

1. Introduction

Artificial intelligence (AI) is fundamentally reshaping economic interactions and business environments (Ma et al., 2025; Xin, 2025; Yang et al., 2025). Firms increasingly announce plans to adopt AI (see Fig. A1), citing gains in efficiency and product quality. However, many such disclosures reflect narrative-driven statements, publicly articulating ambitious future AI strategies without corresponding evidence of technological implementation. In March 2024, SEC Chair Gary Gensler labeled this practice "AI washing." This strategic use of narratives in AI adoption raises a critical question: How do markets respond when firms disclose planned AI adoption absent tangible evidence of actual technological deployment?

The strategic employment of narratives to shape investor perceptions is well-documented (Bertsch et al., 2021; Boitani and Dragomirescu-Gaina, 2023; Liu et al., 2022). Shiller (2020) emphasizes that emotionally salient stories influence market outcomes by directing investor attention. Historical episodes illustrate this mechanism: during the 1970s electronics boom, non-tech firms adopted tech-related names to attract capital (Malkiel, 1999); during the blockchain surge, Long Island Iced Tea rebranded as Long Blockchain Corp., triggering a 200 % stock increase despite having no blockchain activity. The firm was ultimately delisted by the SEC.

Recent breakthroughs in large language models have intensified public and investor attention towards AI, creating incentives for firms to adopt strategically contagious narratives disconnected from implementation. By promoting visionary "Will-Do AI" claims (i.e., statements about planned future AI adoption), firms can capitalize on investor optimism without incurring immediate costs. The complexity and opacity of AI increase the scope for such exaggeration, creating a gap between stated intentions and verifiable actions ("Done AI", i.e., observable AI implementation such as products, patents, or acquisitions). This divergence offers a unique setting to examine the market and firm-level consequences of strategic narrative disclosures.

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This study draws from signaling theory (Spence, 1978, 2002) and narrative economics (Shiller, 2020) to show how markets react when unverifiable AI claims proliferate. Specifically, we collect Form 10-K filings of U.S. publicly listed firms from 2018 to 2023 using the SEC's Electronic Data Gathering, Analysis, and Retrieval (EDGAR) system. Using BERT-based classification and patent evidence, we differentiate speculative "Will-Do AI" from substantive "Done AI". The final model achieves an accuracy of 0.87. We also incorporate patent data from the USPTO and PATSTAT databases to supplement the measurement of "Done AI." To examine market responses, we use BHAR and CAR to capture short-term investor reactions and sales growth to measure the long-term effects of speculative AI disclosures.

The evidence is consistent with our hypothesis: speculative AI disclosures boost short-term attention but ultimately generate negative investor reactions and weaker firm performance. These findings connect to research on AI adoption (Bonney et al., 2024; Haenlein and Kaplan, 2021; Ma et al., 2025; Pisanelli, 2022; Wang et al., 2023; Xin, 2025) and strategic disclosure (Akyildirim et al., 2020; Cheng et al., 2019), documenting a market backlash to "AI washing." The broader implication is that while narratives can mobilize attention, credibility remains essential: markets eventually punish rhetoric that outpaces substance.

The paper is structured as follows. Section 2 presents the literature review and hypotheses development, Section 3 describes the data and research design, Section 4 reports the empirical results, and Section 5 concludes.

2. Related literature and hypotheses

Our analysis builds on signaling theory (Spence, 1978, 2002) and models of cheap talk (Chakraborty and Harbaugh, 2010; Milgrom and Roberts, 1986), which explain why firms facing information asymmetry have incentives to overstate capabilities (Connelly et al., 2011). Exaggerated claims about artificial intelligence are a modern extension of this logic: they are low-cost signals with limited verifiability (Crawford and Sobel, 1982), designed to exploit investor optimism in settings where technological opacity creates room for manipulation.

Yet such narratives do not remain neutral. As transparency increases, markets are better equipped to evaluate rhetoric against evidence. Disclosures framed around future plans rather than achieved outcomes can backfire, signaling weakness rather than strength (Walker and Wan, 2012). This reflects a broader phenomenon: when symbolic communication substitutes for substance, it erodes credibility and reallocates resources away from genuine innovators.

The institutional consequences matter. Symbolic narratives tend to be perceived as opportunistic. Over time, repeated exaggeration undermines trust between firms, investors, and regulators, creating precisely the type of credibility traps that weaken financial markets. We therefore expect markets to penalize inflated AI claims.

H1. Overclaiming AI adoption results in negative investor responses.

The costs extend beyond short-term price reactions. Firms that emphasize narratives without implementation forego the productivity and efficiency gains of real adoption. This rhetoric–reality gap diminishes customer trust, undermines reputation, and slows long-term growth (Kirca et al., 2005; Walker and Wan, 2012). More broadly, a focus on hype diverts resources from research and development, weakening innovation capacity (Tellis et al., 2009). The result is not only weaker firm performance, but also potential misallocation of capital and talent in the wider economy.

H2. Overclaiming AI adoption negatively affects firms' sales performance.

3. Research design

3.1. Data and sample

We collect 10-K filings for U.S. public firms from 2018 to 2023 from the SEC's EDGAR system. To measure AI-related disclosure, we fine-tune a BERT model to distinguish between two dimensions. *Will-Do AI* refers to forward-looking statements about planned adoption, such as: "…We also seek to make acquisitions in technology that enhance our marketing, artificial intelligence and research and development capabilities…" *Done AI* reflects verifiable implementation (e.g., product deployment, specialist recruitment, or AI-related acquisitions), for example: "…The project leveraged machine learning (ML) using big data to serve as the digital twin of the blade manufacturing process…" Table A2 provides additional examples for each category along with the associated firm names and filing dates. For each filing, we identify sentences containing AI-related keywords and apply a semantic filter to ensure that the extracted sentences pertain to AI activity. We construct the training dataset by manually labeling all AI-related sentences in a randomly selected set of 1000 filings from the sample period. The resulting classifier attains an out-of-sample accuracy of 0.87.

To supplement *Done AI*, we integrate patent data from USPTO and PATSTAT to further validate our measurement of Done AI. Following Dyevre and Seager (2023), we address challenges such as firm-name inconsistencies, mergers and acquisitions affecting patent ownership, and the need to assign patents to ultimate parent entities. Drawing on Pairolero et al. (2025), we identify AI-related patents and compute the ratio of AI-related patents to total patents for each firm. This ratio complements our text-based measure and provides an external validation of firms' AI engagement.

Investor reactions are captured by buy-and-hold abnormal returns (BHAR) around 10-K filing dates (Cheng et al., 2019) ,[1] benchmarked to the Fama–French three-factor model (Fama and French, 1993). As a robustness check, we also compute cumulative abnormal returns (CAR).

To control for firm heterogeneity, we include standard covariates from prior studies (Chen et al., 2024; Duchin et al., 2010; Frank and Goyal, 2009; Hussainey and Aal-Eisa, 2009; Mahmud et al., 2022), including size, leverage, earnings growth, debt ratios, market leverage, trade credit, cash flow, quick ratio, tangibility, and Tobin's Q. All continuous variables are winsorized at the 1st and 99th percentiles, depending on the lowest boundary. Summary statistics are reported in Table 1.

3.2. Models

We begin with an event study to assess initial investor reactions to AI overclaiming, measured by buy-and-hold abnormal returns (BHAR) around 10-K filing dates. The sample consists of firms disclosing *Will-Do AI* without evidence of *Done AI*.

To give support to the causal interpretation, we implement a difference-in-differences (DiD) design. AI disclosure is measured as the share of AI-related sentences in each filing and benchmarked against industry-year averages. Firms with above-industry disclosure but below-industry AI patenting are classified as overclaiming. Our DiD specification is:

$$Y_{i,t} = \beta_0 + \beta_1 Treat * Post_{i,t} + \beta_2 Treat_{i,t} + \beta_3 Post_{i,t} + \beta_4 X^{'}_{i,t} + \Psi_i + \eta_t + \varepsilon_{i,t} \tag{1}$$

Where the dependent variable $Y_{i,t}$ represents BHAR and CAR for firm *i* in year *t*. Drawing on Cheng et al. (2019), BHAR is computed as the firm's buy-and-hold return from t1 to t2 minus the buy-and-hold return of a benchmark portfolio. For the main analysis, the benchmark portfolio return from t1 to t2 is estimated using the Fama–French three-factor model with an estimation window of (−175, −26) (Fama and French, 1993). CAR is defined as the cumulative abnormal return over the event window. Abnormal returns are calculated as the firm's actual return minus the expected return predicted by the Fama-French three-factor model (Fama and French, 1993). $Treat_{i,t}$ is an indicator for AI overclaiming and equals one if firm *i* overclaims its AI usage in filing year *t*, and zero otherwise. $Post_{i,t}$ equals one for observations after the release of firm *i*'s annual report in year *t*, which occurs every year, and zero otherwise. Accordingly, the DiD estimates should be interpreted as within–filing-year comparisons around disclosure timing, rather than as a one-time adoption effect.

$X^{'}_{i,t}$ is a vector of firm controls. We also include firm fixed effects, Ψ_i, and year fixed effects, η_t. Errors are two-way clustered by firm and year. Variables are defined in Table A1.

To evaluate real outcomes, we estimate:

$$Sales\ Growth_{i,t} = \beta_0 + \beta_1 Overclaim\ AI_{i,t-1} + \beta_2 X^{'}_{i,t} + \Psi_i + \eta_t + \varepsilon_{i,t} \tag{2}$$

where $Overclaim\ AI_{i,t-1}$ is the lagged binary indicator identifying overclaiming firms, defined as firms whose public claims of AI adoption exceed their verifiable technological implementation. To address endogeneity, we instrument with the industry-average overclaiming rate within four-digit SIC codes (Ioannou et al., 2023). Our instrument, the industry-average rate of AI overclaiming within four-digit SIC codes, captures the interdependence of disclosure strategies across firms. Managers often imitate peers in shaping disclosure narratives, which generates strong cross-firm correlation in overclaiming behavior. At the same time, these peer disclosure patterns are unlikely to directly influence a focal firm's sales outcomes. We acknowledge that industry-level shocks could potentially affect both peer disclosure behavior and firm sales, which may threaten the exclusion restriction. While this approach cannot fully rule out all sources of correlated shocks, it follows standard practice in the literature. To partially address this concern, we use a leave-one-out industry average as the instrument, which helps mitigate concerns about direct mechanical correlation.

4. Empirical results

4.1. Event study

Our treatment group, the overclaiming sample, comprises firms that disclose Will-Do AI without implementation evidence. Sample firms are selected based on two criteria: (1) whether the firm discloses forward-looking statements indicating strategic plans, commitments, or expectations regarding future AI adoption, and (2) the extent to which the disclosed AI strategy is supported by observable implementation, as measured by AI-related patents. To mitigate concerns about industry-specific concentrations in AI strategy, firms exhibiting above-industry levels of forward-looking AI disclosure but below-industry levels of AI patenting are classified as overclaiming.

As shown in Fig. 1, the event study results indicate that overclaiming firms initially experience a short-lived positive market

[1] We use BHAR as the main measure of market reaction for two reasons. First, our analysis focuses on whether price reactions persist or reverse after AI disclosure. BHAR naturally captures the path of investors' realizable holding-period returns from the event window to the post-event window, making it well-suited for testing short-term overreaction and subsequent price reversals. Second, AI disclosures often exhibit speculative characteristics. BHAR is more effective at capturing extreme price increases because it incorporates compounding and the return path.

Table 1
Summary statistics.

VARIABLES	N	Min	Median	Mean	SD
Tobin's Q	11,320	−0.357	1.359	4.626	21.209
Leverage	11,320	0.003	29.902	42.697	79.182
Tangibility	11,320	0.000	0.158	0.248	0.235
Quick Ratio	11,320	0.004	1.372	2.416	3.369
Sales Growth	11,320	−1.000	0.062	0.256	1.115
Earnings Growth	11,320	−10.831	0.015	−0.012	2.368
Net Trade Credit	11,320	0.000	0.089	0.115	0.107
Market Leverage	11,320	0.000	0.186	0.256	0.239
Operating Cash Flow Ratio	11,320	−9.692	0.236	−0.090	1.567
Long Debt Ratio	11,320	0.000	0.887	0.774	0.271
Size	11,320	0.014	6.691	6.446	2.494
Treat	11,320	0.000	0.000	0.063	0.244
Percentage of AI-Related Sentences	11,320	0.000	0.000	0.000	0.001
AI Literacy	11,320	0.000	0.000	0.171	0.377

response (days −10 to 0), with BHAR reaching approximately 2.29 %. However, this optimism fades significantly by day +1, with returns declining to 1.81 % and becoming statistically insignificant ($p > 0.10$) beyond day +2, as the 95 % confidence intervals cross zero. This pattern suggests that while investors may initially reward AI aspirations, they eventually penalize firms for lacking substantive AI implementation, consistent with the market efficiently distinguishing between credible and non-credible AI claims. The eventual return reversal implies market overreaction correction.

We classify firms by their stated AI adoption strategy. As shown in Fig. 2, firms pursuing AI through partnerships or acquisitions see stronger initial market reactions than those relying on internal R&D, likely because such strategies are more verifiable. Similarly, firms disclosing AI use for operational efficiency experience larger early gains than those focused on product enhancement (see Fig. A2). In all cases, the effects fade and become statistically insignificant after 30 days.

Building on the event study results, we implement a more rigorous difference-in-differences (DiD) analysis to isolate the causal effect of AI overclaiming. Table 2 presents the results from Model (1). The consistently negative and statistically significant coefficients both on CAR and BHAR across all specifications are consistent with market penalties associated with AI overclaiming. Specifically, columns (2) and (4) of Table 2 indicate that if a firm engages in AI overclaiming in its 10-K filing, the release of the 10-K is followed by a 1.6 % decline in CAR and a 1.4 % decline in BHAR, respectively. This result is robust to including firm-by-year fixed effects, as presented in Table A3. The main DiD analysis in Table 2 uses a shorter event window of [−10, +10] to focus on immediate market responses around disclosure. As a robustness check, we extend the window to [−10, +30] to capture any delayed price adjustments, as reported Table A4. These results confirm that investors distinguish between credible and non-credible AI disclosures, rewarding only those backed by tangible implementation evidence. The economic magnitude suggests material market penalties for strategic AI overclaiming.

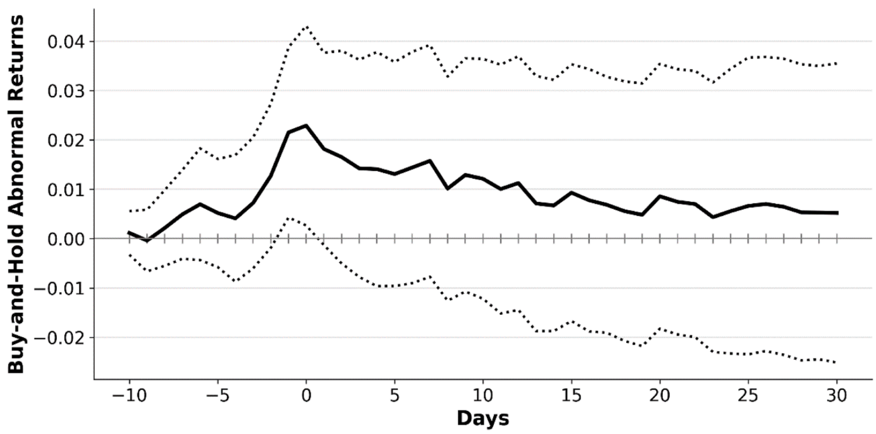

Fig. 1. Investor reactions to the overclaiming AI adoption of treatment firms.
Notes: This figure plots the BHAR (relative to the Fama-French three-factor model) from ten days before to thirty days after the date of 10-K filings for firms overclaiming AI adoption. The solid lines represent the BHAR returns, and the dotted lines represent 5 %/95 % confidence intervals.

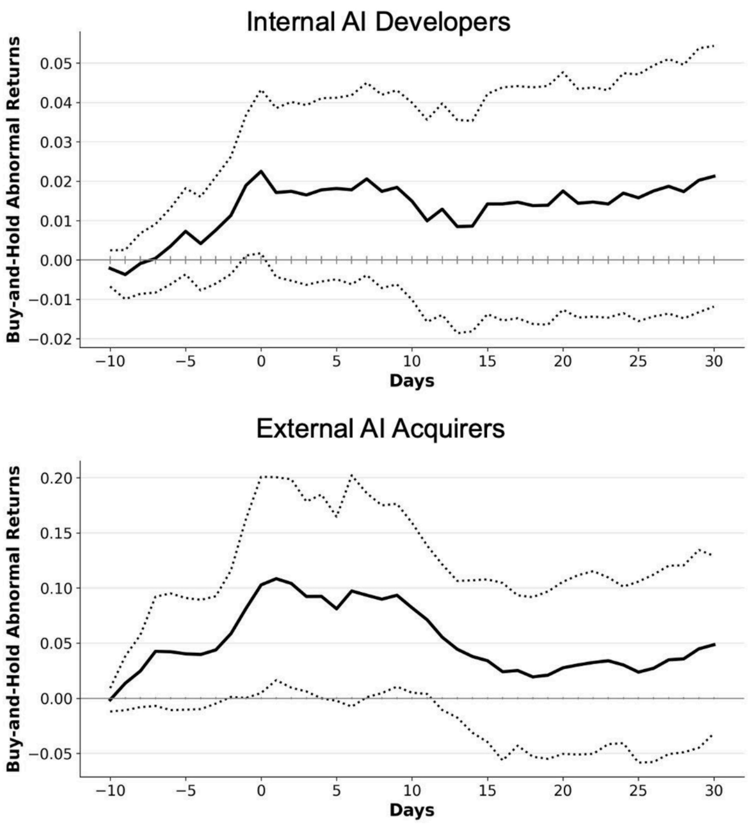

Fig. 2. Investor reactions to the overclaiming AI adoption by strategic approach.
Notes: This figure plots the BHAR (relative to the Fama-French three-factor model) from ten days before to thirty days after the date of 10-K filings for firms overclaiming AI adoption. Internal AI Developers are firms claiming they will develop AI capabilities through their own research and development activities. External AI Acquirers are firms stating they will adopt AI through acquisitions of AI-related firms or technologies, or through partnerships with AI-focused companies. The solid lines represent the BHAR returns, and the dotted lines represent 5 %/95 % confidence intervals. Classification is based on a fine-tuned BERT model, as previously described.

4.2. Financial performance

To establish the real economic consequences of AI overclaiming, we analyze its impact on subsequent sales growth using a rigorous empirical framework. The results of Model (2), presented in Table 3, reveal statistically and economically significant negative effects: firms engaging in AI overclaiming experience substantially lower sales growth in subsequent periods compared to their peers with verified AI implementation. Take column (2) as an example: overclaiming AI adoption would reduce sales growth in the following year by 14.7 percentage points. Columns (3) and (4) report the IV-2SLS estimates, using the industry-average rate of overclaiming AI adoption as the instrumental variable, and the results remain consistent with the OLS estimates. In Column (5), we employ an alternative instrument based on the leave-one-out industry average as an additional robustness check. The resulting estimate is consistent with the baseline but larger in magnitude, potentially reflecting weaker first-stage strength after excluding the firm's own observation. Taken together, the results align with the interpretation that the negative market reaction to unsubstantiated AI claims may coincide with deteriorating firm performance, although we acknowledge that endogeneity cannot be fully ruled out due to the potential correlation with the industry-level shocks.

We further assess whether higher AI-related information asymmetry exacerbates or mitigates the negative consequences of overclaiming AI capabilities. We collect state–year AI search indices for the United States from 2018 to 2023 using Google Trends. We define AI literacy as a dummy variable equal to 1 if a state's search intensity in a given year exceeds the national average for that year, and 0 otherwise. Firms are then matched to state-level AI literacy based on the location of their headquarters, and an interaction term is added to Eq. (2). As shown in Table A5, the penalty for Overclaim AI is most severe in states with the highest search activity, where public AI literacy is strongest.

Table 2
Difference-in-difference test.

	CAR		BHAR	
VARIABLES	(1)	(2)	(3)	(4)
DiD	−0.0159**	−0.0160**	−0.0139**	−0.0140**
	(0.005)	(0.005)	(0.005)	(0.005)
Treat	−0.0028	−0.0049	−0.0024	−0.0046
	(0.010)	(0.009)	(0.010)	(0.009)
Post	−0.0008	−0.0007	−0.0006	−0.0005
	(0.003)	(0.003)	(0.003)	(0.003)
Observations	20,330	20,330	20,330	20,330
R-squared	0.212	0.219	0.215	0.222
Firm Controls	YES	YES	YES	YES
Firm FE	YES	YES	YES	YES
Year FE	NO	YES	NO	YES
Standard Errors	Clustered	Clustered	Clustered	Clustered

Notes: This table reports the DiD results of CAR and BHAR. The dependent variables are CAR and BHAR over the [−10, +10] trading day window surrounding the event date, respectively. The key explanatory variable, DiD, is an interaction term between Treat (an indicator for firms announcing AI without substantial adoption) and Post (an indicator for the post-announcement period). Firm controls include firm size, leverage, earnings growth, long-term debt ratio, market leverage (debt-to-enterprise value), net trade credit, operating cash flow ratio, quick ratio, asset tangibility, and Tobin's Q. Columns (1)-(2) report cumulative abnormal returns (CAR) and columns (3)-(4) show buy-and-hold abnormal returns (BHAR). All specifications are two-way clustered based on the firm and year. See Table A1 for more detailed definitions of the variables and data sources. *, ** and *** indicate significance at the 10 %, 5 %, and 1 % levels, respectively.

Table 3
Overclaiming AI adoption & sales growth.

	Sales Growth				
VARIABLES	(1)	(2)	(3)	(4)	(5)
L.Overclaim AI	−0.176***	−0.147***	−0.178*	−0.172**	−0.921**
	(0.027)	(0.027)	(0.091)	(0.081)	(0.417)
Observations	10,865	10,865	11,320	11,320	11,073
R-squared	0.382	0.393	0.024	0.034	0.005
F-stats	.	.	68.398	68.270	3.594
Firm Controls	YES	YES	YES	YES	YES
Year FE	NO	YES	NO	YES	YES
Specification	OLS	OLS	IV	IV	IV
Standard Errors	Clustered	Clustered	Clustered	Clustered	Clustered

Notes: This table reports the OLS and IV-2SLS regression results of sales growth on lagged overclaim AI. The dependent variable is the sales growth. The key explanatory variable, *Overclaim AI*, is a binary indicator that captures instances where firms' public claims of AI adoption exceed their verifiable technological implementation. The instrument variable is the industry-average rate (four-digit SIC code) of *Overclaim AI* for columns (3) and (4), and the leave-one-out industry average rate for column (5). Firm controls include firm size, leverage, earnings growth, long-term debt ratio, market leverage (debt-to-enterprise value), net trade credit, operating cash flow ratio, quick ratio, asset tangibility, and Tobin's Q. Firm fixed effects are controlled for all OLS regressions. All specifications are two-way clustered based on the firm and year. See Table A1 for more detailed definitions of the variables and data sources. *, ** and *** indicate significance at the 10 %, 5 %, and 1 % levels, respectively.

5. Conclusion

This study shows that when firms exaggerate their adoption of artificial intelligence, markets eventually impose discipline. Initial investor enthusiasm gives way to skepticism once it becomes clear that claims lack substantive technological implementation. In this sense, AI overclaiming represents a modern form of cheap talk: narratives that mobilize attention but, when disconnected from reality, erode credibility and impose long-term costs. This finding extends insights from signaling theory (Spence, 1978, 2002) and narrative economics (Shiller, 2020) to the study of AI adoption and disclosure (Wang et al., 2023), highlighting the potential backlash associated with exaggerated AI claims.

The implications extend beyond firm performance. History shows that cycles of overstatement, whether in electronics in the 1970s, internet firms in the late 1990s, or blockchain in the 2010s, distort the allocation of capital and weaken trust in markets. Our evidence suggests that the same pattern is emerging in AI. Our findings suggest that, if left unchecked, “AI washing” not only damages individual firms but risks diverting resources away from genuine innovators, reducing productivity growth in the broader economy.

For regulators, these results underscore the importance of disclosure standards that match the complexity of new technologies. For investors, they illustrate that hype is not a substitute for implementation. And for managers, they serve as a warning: while narratives can buy temporary attention, sustainable value creation requires substance. Markets, over time, punish rhetoric that outpaces reality.

This study is not without limitations. Due to the temporal scope of available data, the identification strategy, while designed to provide causal inference, cannot fully eliminate concerns about unobserved technological developments that may materialize over longer horizons. Future research would benefit from richer, more granular data over an extended period, which could generate cleaner causal estimates and further validate the robustness of these findings across industries and market conditions.

Author statement

We hereby confirm the following:

Originality

This manuscript is original, has not been published previously, and is not under consideration for publication elsewhere.

Ethical approval

This study does not involve human subjects, animal experiments, or sensitive personal data; therefore, ethical approval is not required.

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CRediT authorship contribution statement

Xiapeng Song: Writing – original draft, Software, Methodology, Conceptualization. **Wenxuan Hou:** Writing – review & editing, Supervision. **Zizhou Ouyang:** Validation, Data curation. **Fangmin Hao:** Writing – original draft, Methodology, Formal analysis.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:10.1016/j.frl.2026.109684.

Data availability

Data will be made available on request.

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